

1. KICAD 9 PCB WORKFLOW

1. **Schematic (.kicad_sch)**: Draw components & electrical connections.
2. **Assign Footprints**: Pair schematic symbols with physical footprints.
3. **Run ERC**: Pass Electrical Rules Check (fix floating pins & nets).
4. **Update PCB (F8)**: Sync component changes into PCB Editor.
5. **Board Layout (.kicad_pcb)**: Draw **Edge.Cuts** boundary & route tracks.
6. **Run DRC**: Pass Design Rules Check (fix clearance/overlap issues).
7. **3D View (Alt + 3)**: Inspect physical geometry & graphics.

2. VIEW & NAVIGATION

Scroll Wheel	Zoom in / Zoom out
Middle Drag / Right Drag	Pan Canvas View
Esc	Cancel active tool / Deselect item
Spacebar	Reset relative coordinate origin (0,0)
Alt + 3	Open 3D Viewer window

3. SCHEMATIC EDITOR (EESCHEMA)

A	Add Symbol (components, power, GND)
W	Wire (draw electrical connections)
R	Rotate component (90° clockwise)
M	Move item (disconnects attached wires)
G	Grab item (moves item, keeping wires attached)
C	Duplicate / Copy symbol under cursor
E	Edit Properties (change value, e.g., 220Ω)
Del / Backspace	Delete selected component or wire

KiCad 9 Tip: Hover over any symbol or track and press hotkeys directly without clicking first!

4. PCB LAYOUT EDITOR (PCBNEW)

F8	Sync changes from Schematic to PCB
X	Route Track (draw copper trace)
V	Add Via (switch trace between copper layers)
F	Flip component side (Top ↔ Bottom)
B	Re-fill copper zones & ground planes
Ctrl + M	Measure distance tool
PgUp / PgDn	Switch active working layer (Front / Back)

5. KEY LAYERS COLOR CODE

F.Cu (Red)	Front Copper layer (traces & pads)
B.Cu (Blue)	Back Copper layer (bottom traces)
F.Silks (Cyan)	Front Silkscreen (text, logos, art)
Edge.Cuts (Yellow)	Physical board outline shape
F.Mask (Purple)	Soldermask opening (exposes pads)

6. PRE-FAB CHECKLIST & TROUBLESHOOTING

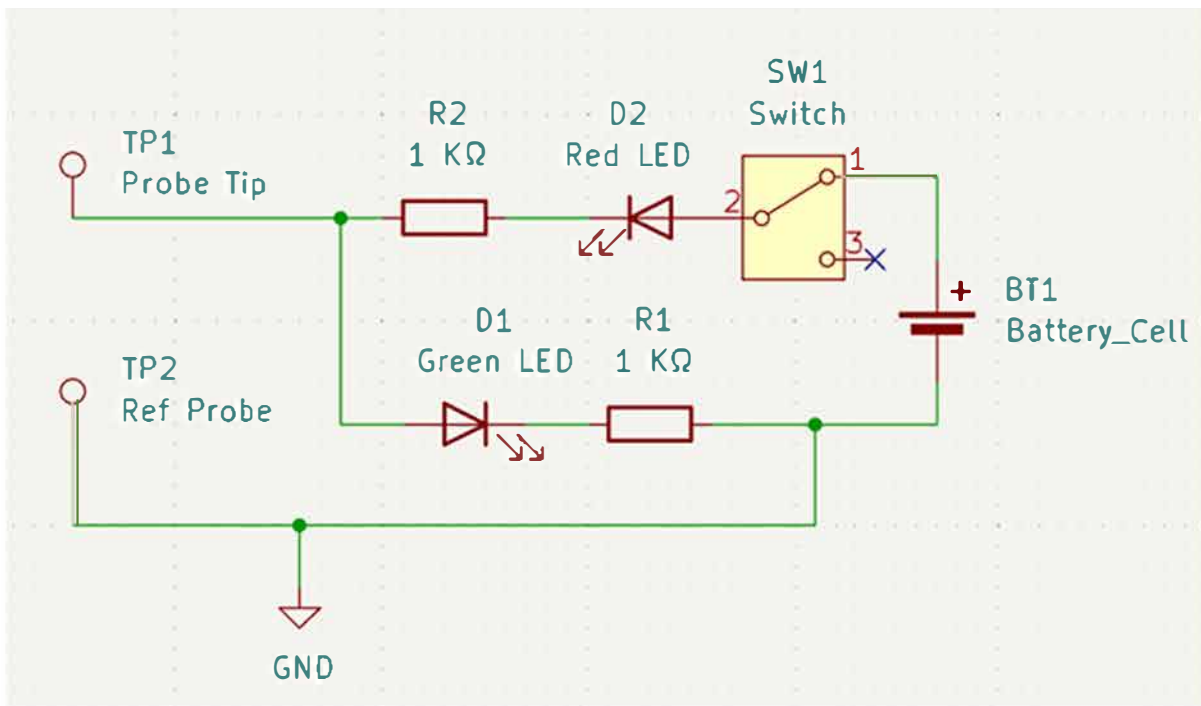
- **Unconnected Pins:** Attach a *No-Connect flag* (**Q**) to intentionally unused pins during schematic capture.
- **Edge.Cuts Boundary:** Ensure the yellow outline is a *100% closed loop* with zero gaps or intersecting lines.
- **Silkscreen Text:** Keep text clear of exposed pads; keep text height ≥ 1.0 mm for legible printing.
- **DRC Check:** Must have 0 errors & 0 warnings before exporting fabrication files!
- **Export Step:** File → Fabrication Outputs → Gerbers (.gbr) & NC Drill (.drl) → Zip output folder.

Soldering Prep: Ensure **thermal reliefs** are enabled on Ground pads so heat isn't drawn away during hand-soldering

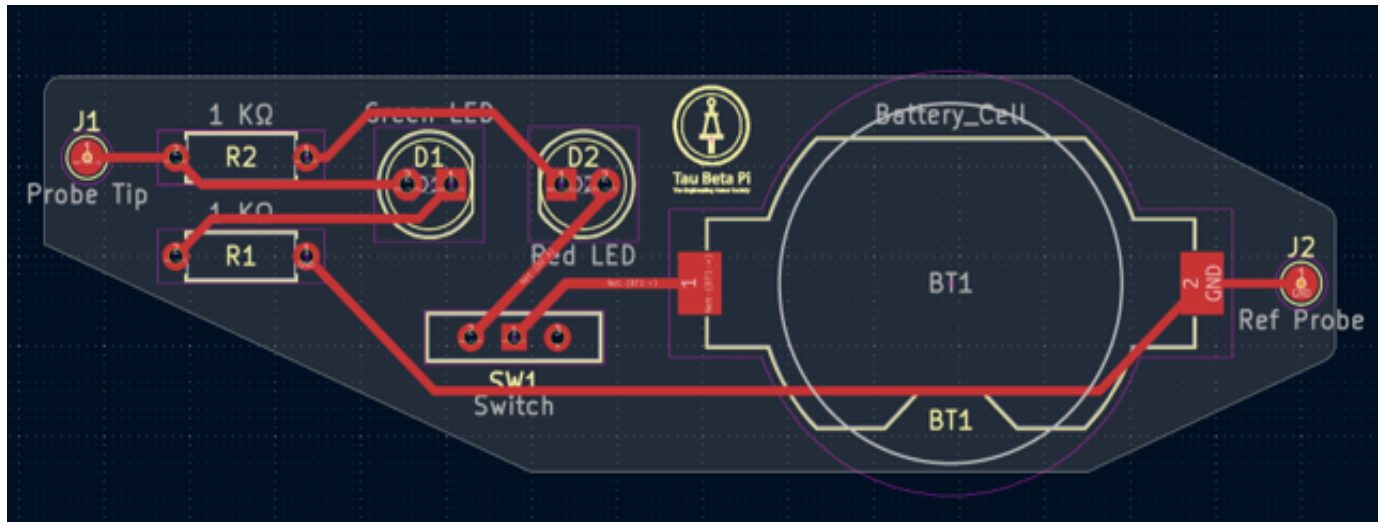
Table of Materials

Reference	Component	Symbol	Footprint
BT1	CR2032 3V Battery Holder	Battery_Cell	BatteryHolder_Keystone_1058_1x2032
SW1	SPDT Slide Switch	SW_SPDT	SW_Slide-03_Wuerth-WS-SLTV_10x2.5x6.4_P2.54mm
D1, D2	Green, Red LED	LED	LED_D5.0mm
R1, R2	1KΩ Resistor	R	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
TP1, TP2	Signal Probe Tip, GND Ref Terminal	TestPoint	TestPoint_Pad_D2.0mm

Schematic Template



PCB Editor Template



3D Image Viewer

